AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 8 · WORKBOOK

Operational Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected operational-label record.

Apply the calendar guide, replace vague instructions, inspect disagreement and retain a policy change.

Keep one habit: Keep field evidence, individual labels, guide version and correction reasons together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Rows Columns and Data Types

Loops and Stopping

Decompose a Goal

W1 · Apply the ordered calendar guide

> Cite a criterion for every label.

Your evidence record

- Calendar guide version 1: whole-number days 1–30 inclusive; room A or B. Any known failure selects revise first. Otherwise any blank or unreadable required field selects review. Otherwise select ready. A label checks printed fields; it does not confirm a booking or grant permission.

- The calendar policy accepts whole-number days 1 through 30 inclusive and a room code A or B. Check known failures first, then unknown fields, then readiness.

  Label these cards and give **evidence**: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Replace looks nice with observable checks

> Include unknown fields and rule priority.

Your evidence record

- Calendar guide version 1: whole-number days 1–30 inclusive; room A or B. Any known failure selects revise first. Otherwise any blank or unreadable required field selects review. Otherwise select ready. A label checks printed fields; it does not confirm a booking or grant permission.

- A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

  Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Distinguish matches from correct applications

> Say whether you used only label lists or also the source cards.

Guide, lists and evidence limit

- Calendar guide version 1: whole-number days 1–30 inclusive; room A or B. Any known failure selects revise first. Otherwise any blank or unreadable required field selects review. Otherwise select ready. A label checks printed fields; it does not confirm a booking or grant permission.

- For the same four cards, Mina records ready, revise, review, ready. Theo records ready, review, review, revise. How many positions agree? Identify the **disagreement**s.

  Can you decide which **annotator** is correct from these **label** lists alone? Explain what else you need.

- Count matches from the two supplied lists first, without consulting earlier cards. From lists alone, correctness is undetermined. If you then explicitly match these positions to W1's supplied cards, you can check against that evidence: Mina differs at position 4 and Theo differs at position 2. State which evidence you used rather than treating agreement as a correctness key.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Keep a changed policy version visible

> An identical record can receive different labels under different definitions.

Your evidence record

- Calendar guide version 1: whole-number days 1–30 inclusive; room A or B. Any known failure selects revise first. Otherwise any blank or unreadable required field selects review. Otherwise select ready. A label checks printed fields; it does not confirm a booking or grant permission.

- Version 1 accepts calendar days 1 through 30. Version 2 accepts days 1 through 31. A card says day 31 room A. State its **label** under each version.

  Explain why mixing the two **label**s without recording the version could create a misleading **training** dataset.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Changed independent case

> Keep one first attempt before hints or feedback.

Read and explain independently

Two **label**lers use book-length rules: fewer than 120 pages and at most 120 pages. **Label** books with 110,120,121 pages. The agreed policy is at most 120. Repair the **disagreement**.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Define 'good match' before **training** or evaluating the recommender.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Replace a vague instruction for Categories and operational **label**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Two **label**ers use 'short' to mean pages<100 and pages<=100. They disagree only on 100-page books. Write a shared definition and explain how to repair the existing **label**s.

▼ Extra practice · A complete case

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Short means fewer than 100 pages. Two **annotator**s disagree about a 100-page book.

Resolve the **label**.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Use a clear **label** rule: short means 80 pages or fewer. Two judges **label** an 80-page book differently. What shared instruction resolves this **disagreement**?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 8 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Two **label**ers use 'short' to mean pages<100 and pages<=100. They disagree only on 100-page books. Write a shared definition and explain how to repair the existing **label**s.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Extra practice · Equipment-request policy

REFERENCE · SECTION 15

Extra practice · Equipment-request policy

> This is a different optional transfer case, not a duplicate of the selected W4.

Apply the fully specified new policy

- A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

  Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

  Explain whether ready alone grants permission to borrow.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

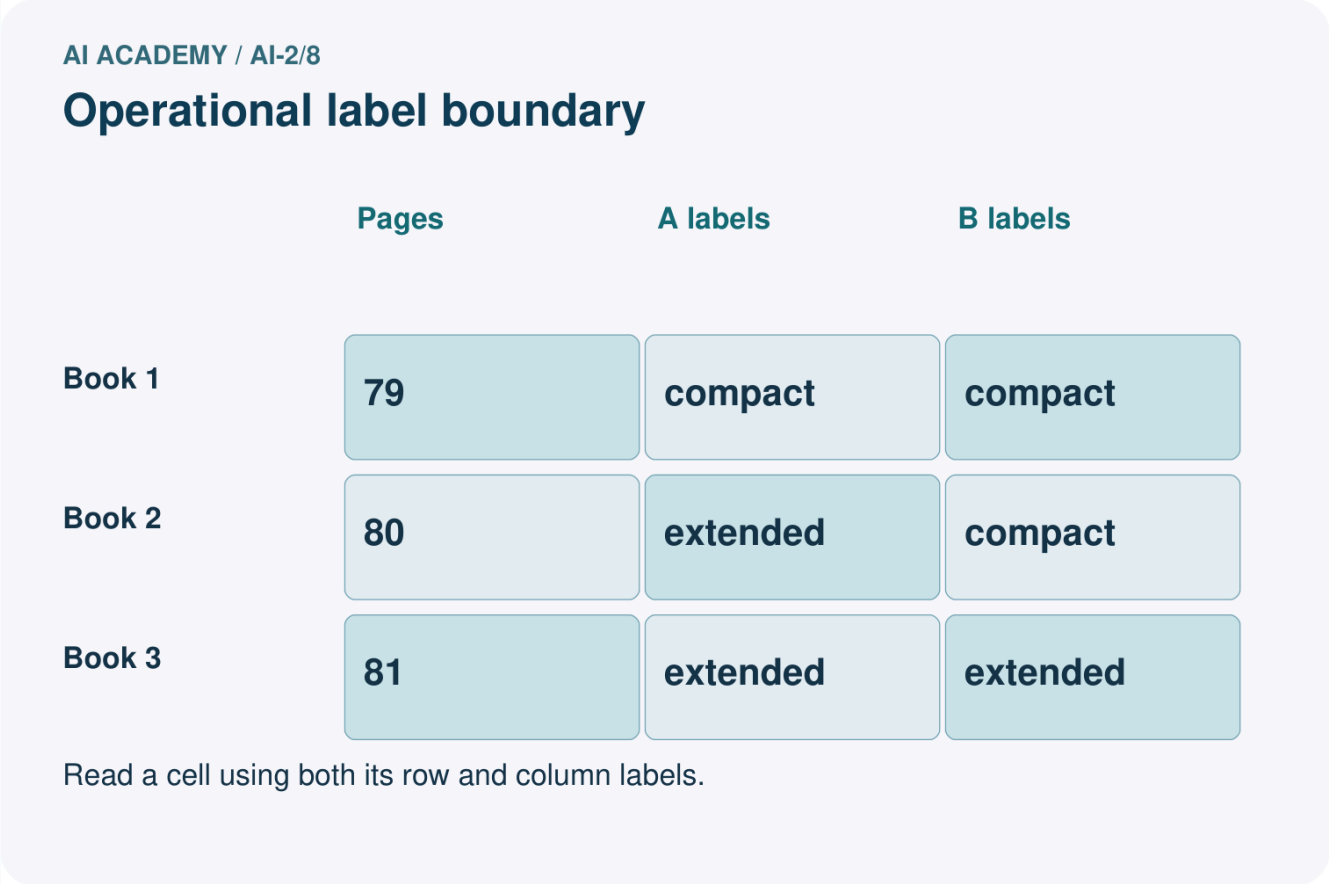
> Keep the explanation and the support you used.

Use this when helpful

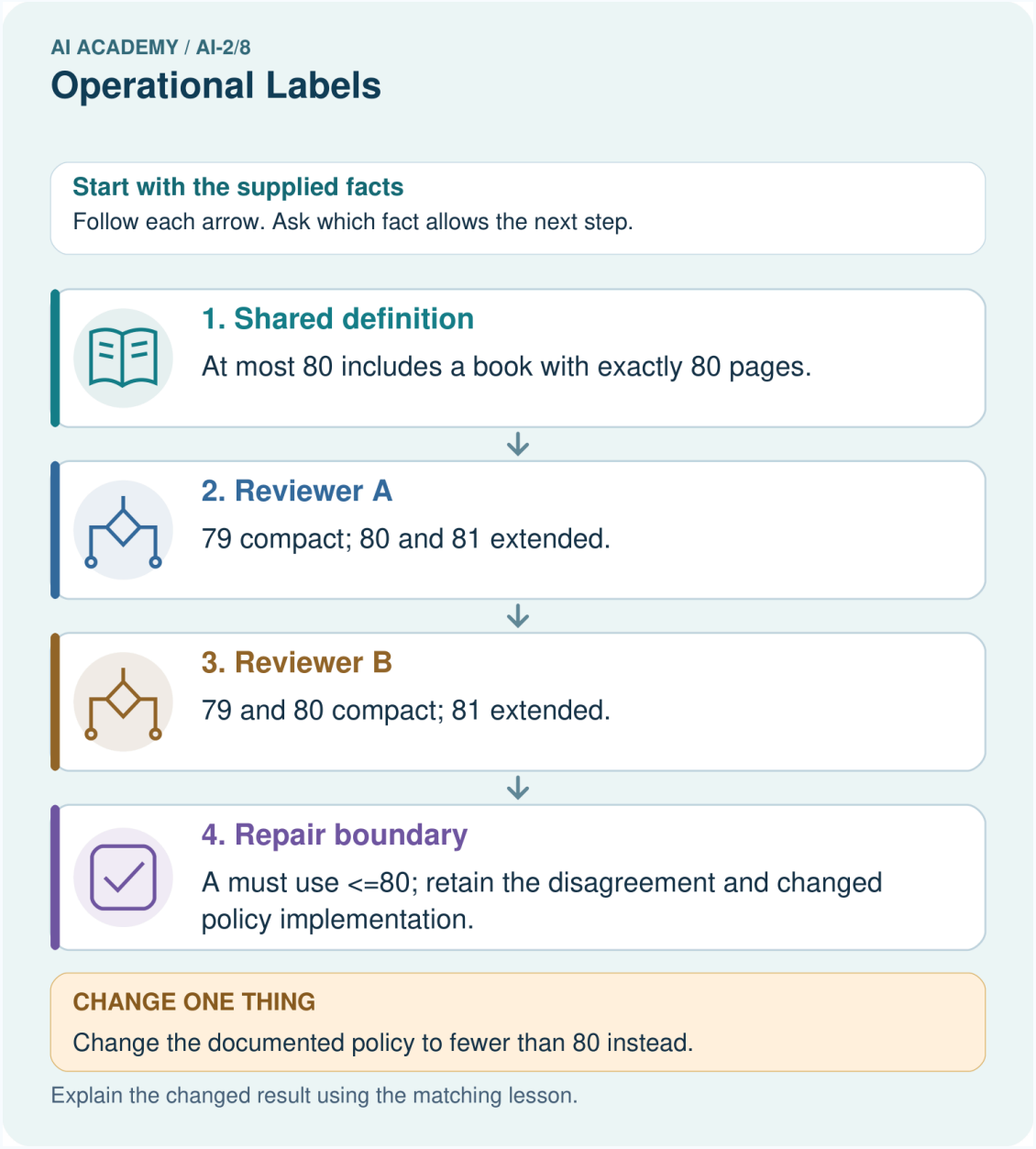
Two fictional catalogue reviewers **label** books compact or extended.

Agreed policy: compact means at most 80 pages; extended means more than 80.

Reviewer A uses pages<80; reviewer B uses pages<=80. Inspect books of 79,80,81 pages.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Two fictional catalogue reviewers **label** books compact or extended.
- Which **label**led step matches this detail? A must use <=80; retain the **disagreement** and changed policy implementation.
- 1. Repair boundary
- 2. Reviewer B
- 3. Reviewer A
- 4. Shared definition
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Repair boundary A must use <=80; retain the **disagreement** and changed policy implementation.

The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions.

Changed-case question

- Changed-case practice: Change the documented policy to fewer than 80 instead. Which conclusion is supported by the supplied facts?
- 1. A's original **label**s now match; B must change its equality behavior.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: A's original **label**s now match; B must change its equality behavior. The authoritative definition changed, not the physical books.

Repair and save your work

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

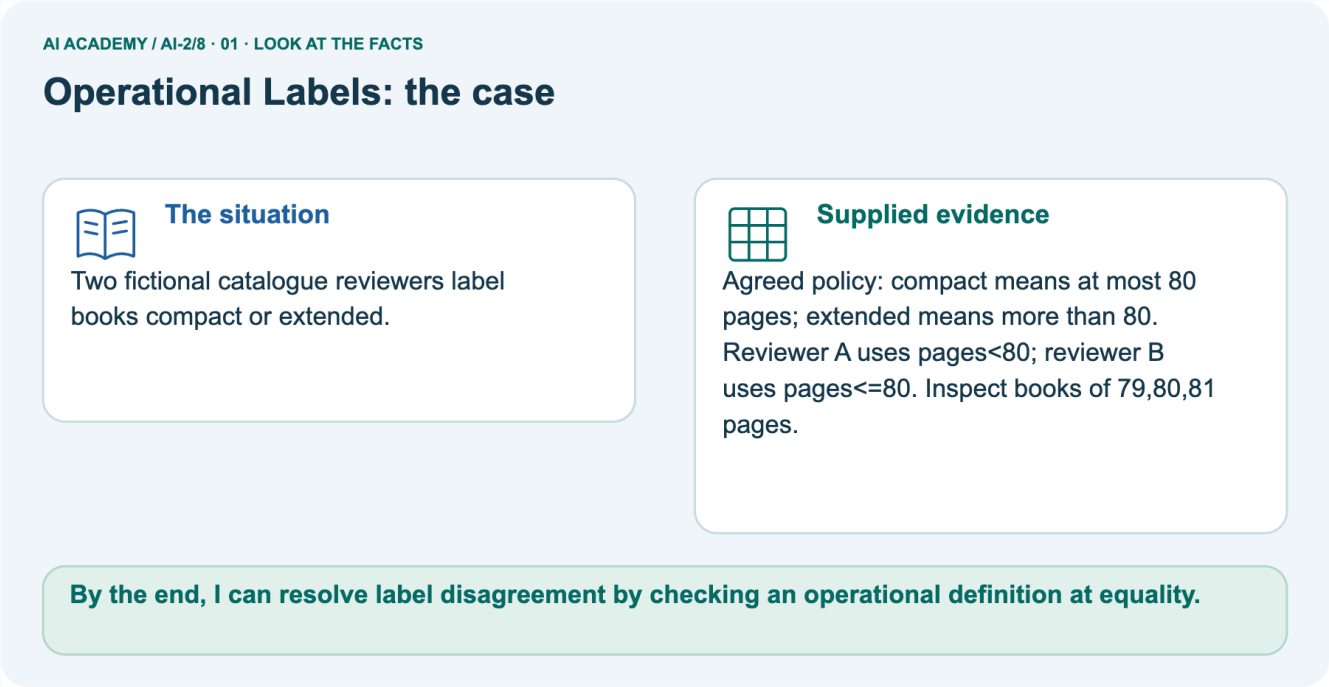
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Operational Labels: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

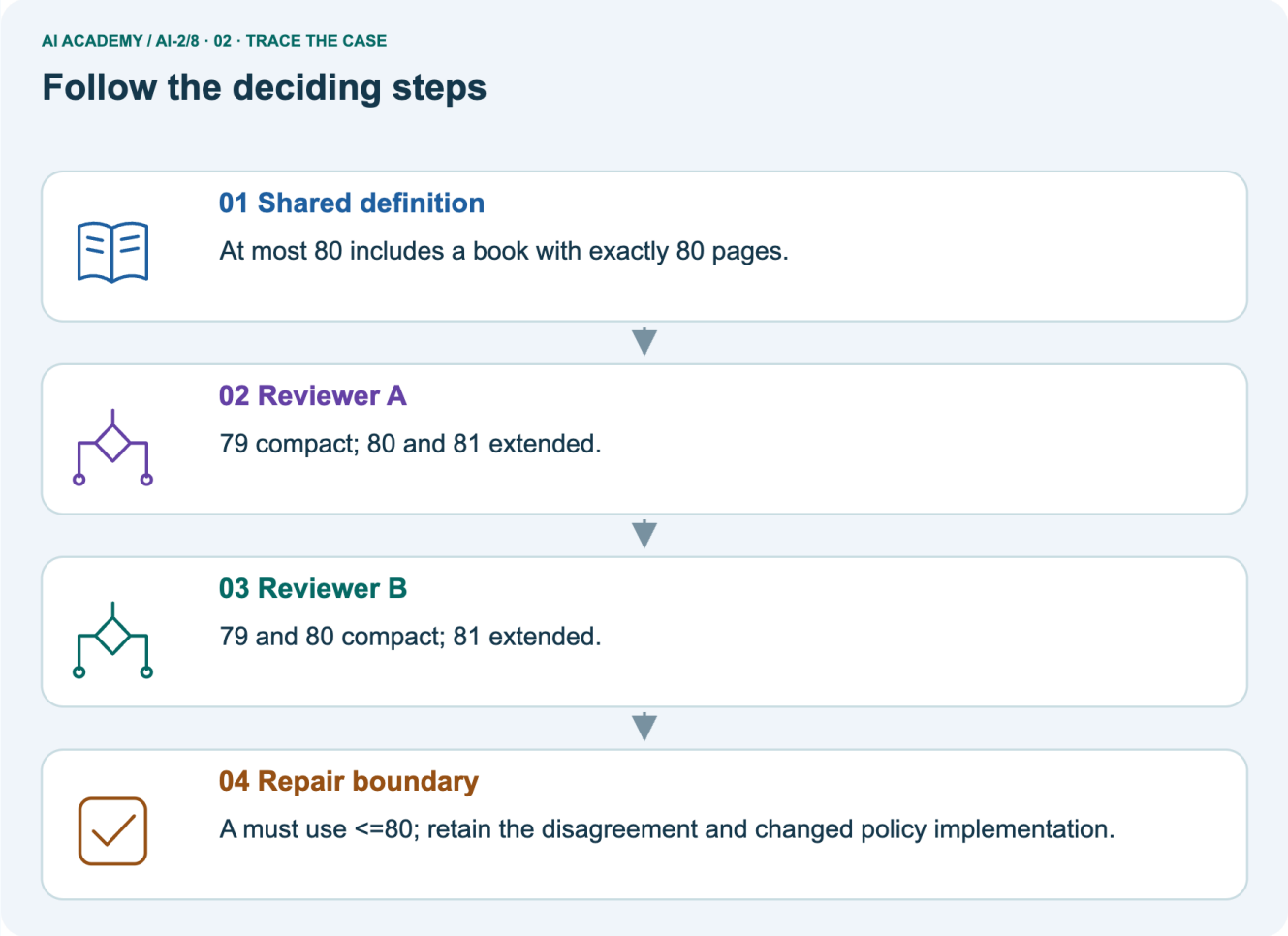
Two fictional catalogue reviewers **label** books compact or extended. Supplied facts: Agreed policy: compact means at most 80 pages; extended means more than 80. Reviewer A uses pages<80; reviewer B uses pages<=80.

Inspect books of 79,80,81 pages.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



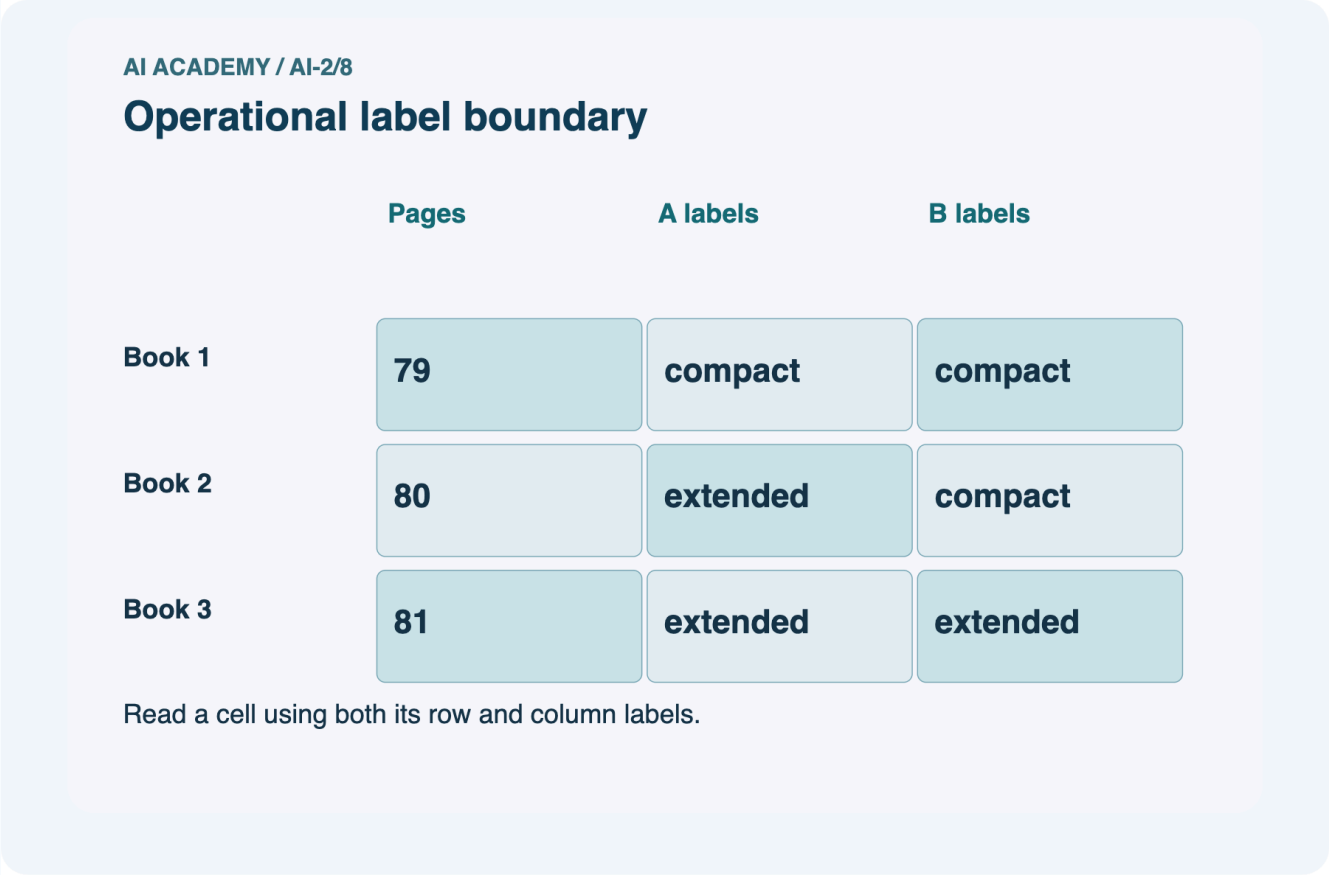
Follow the labelled picture with the matching explanation. Open the vector diagram

Shared definition: At most 80 includes a book with exactly 80 pages. → Reviewer A: 79 compact; 80 and 81 extended. → Reviewer B: 79 and 80 compact; 81 extended. → Repair boundary: A must use <=80; retain the **disagreement** and changed policy implementation.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Operational label boundary



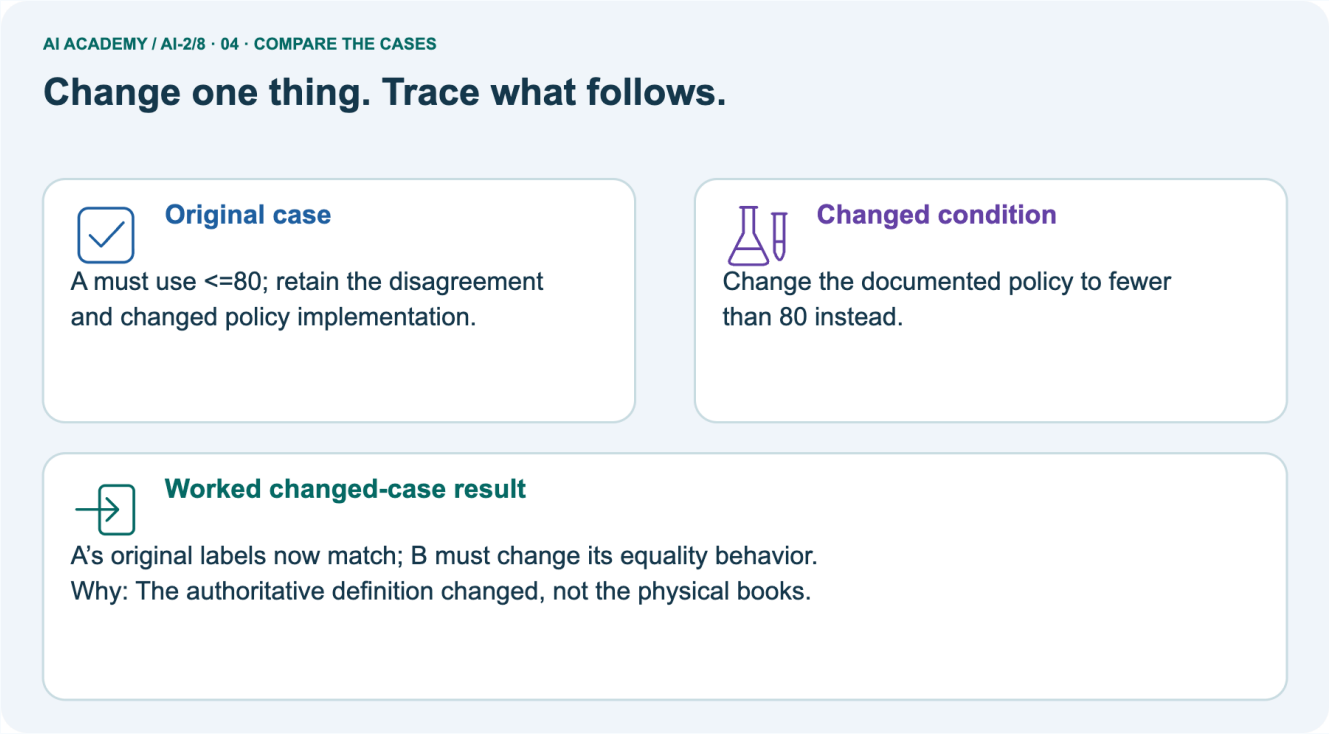
Follow the labelled picture with the matching explanation. Open the vector diagram

Operational **label** boundary. Book 1: Pages = 79, A **label**s = compact, B **label**s = compact. Book 2: Pages = 80, A **label**s = extended, B **label**s = compact. Book 3: Pages = 81, A **label**s = extended, B **label**s = extended.

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

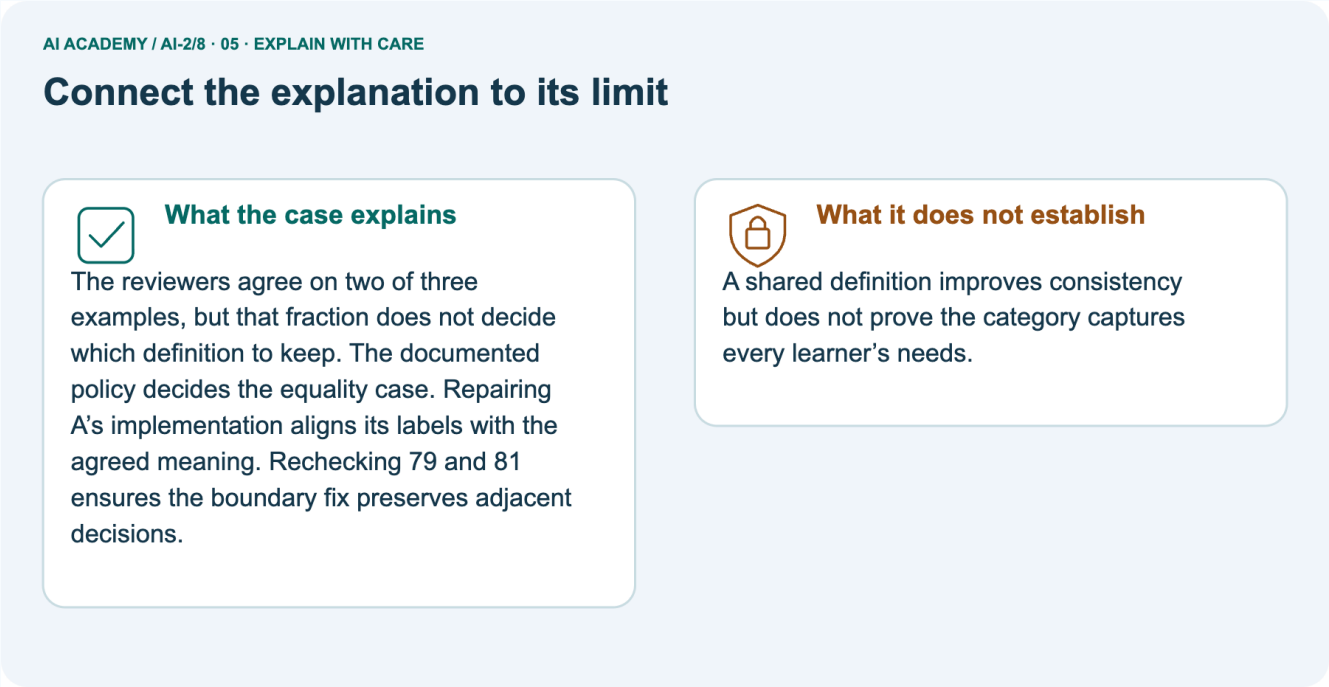
Original case: A must use <=80; retain the **disagreement** and changed policy implementation. Changed condition: Change the documented policy to fewer than 80 instead.

Expected result: A's original **label**s now match; B must change its equality behavior. Reason: The authoritative definition changed, not the physical books.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

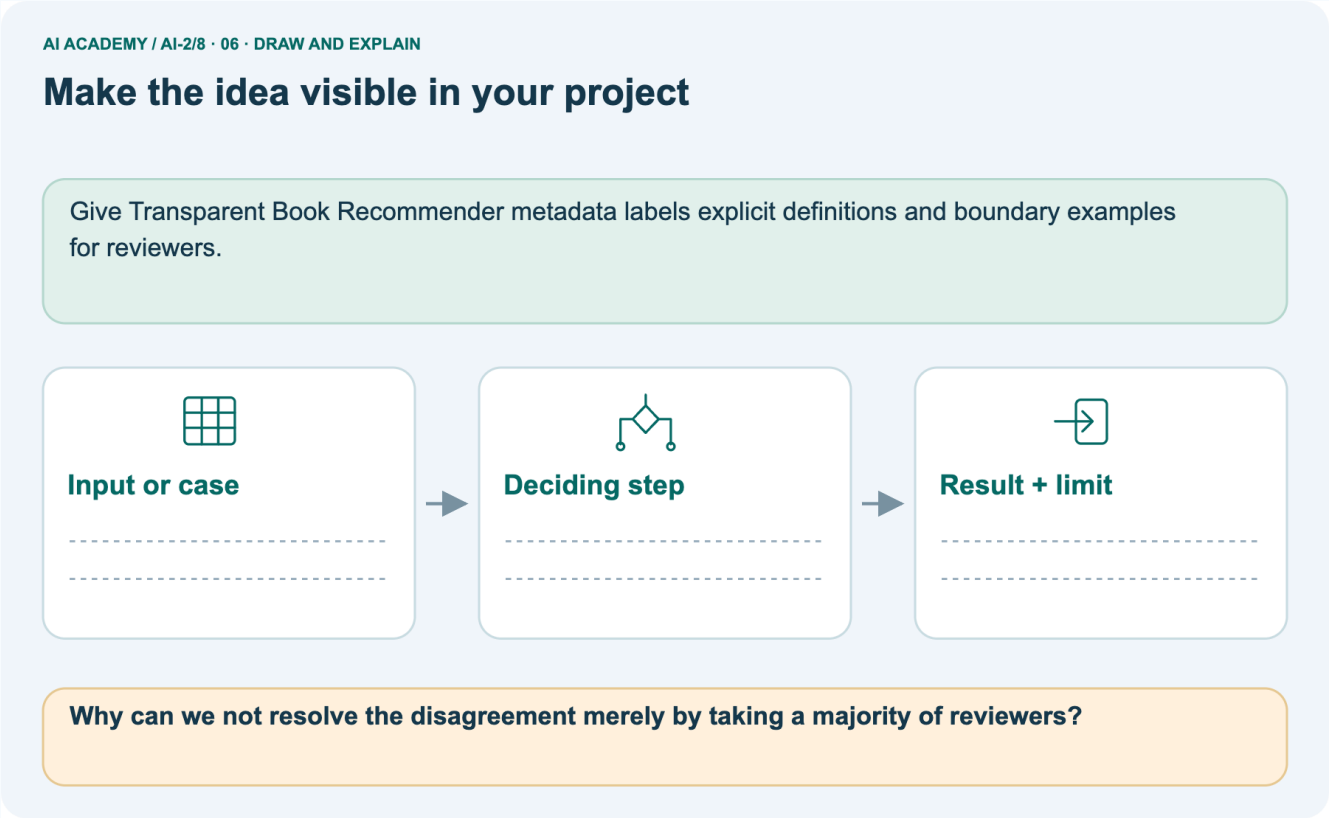
The reviewers agree on two of three examples, but that fraction does not decide which definition to keep. The documented policy decides the equality case. Repairing A's implementation aligns its **label**s with the agreed meaning.

Rechecking 79 and 81 ensures the boundary fix preserves adjacent decisions. Limit: A shared definition improves consistency but does not prove the **category** captures every learner's needs.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give Transparent Book Recommender metadata **label**s explicit definitions and boundary examples for reviewers. Draw your **input**, deciding step, result and limit.

Supported question: Why can we not resolve the **disagreement** merely by taking a majority of reviewers?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Operational Labels: the case

02. Follow the deciding steps

03. Operational label boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.